

Quiz 12

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given a matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$.

(a) Find the eigenvalues λ 's and corresponding eigenvectors x 's of A .

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) - 2 \cdot 5$$

$$= 4 - 5\lambda + \lambda^2 - 10 = \lambda^2 - 5\lambda - 6 = 0$$

$$\Rightarrow (\lambda - 6)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 6, \lambda_2 = -1$$

$$\text{For } \lambda_1 = 6, \text{ solve } (A - 6I)x_1 = 0, \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 = c \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\text{For } \lambda_2 = -1, \text{ solve } (A + I)x_2 = 0, \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_2 = d \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(b) Write an eigenvalue diagonalization $A = SAS^{-1}$ based on the results from (a).

$$S = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}, S^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} = \begin{bmatrix} 6 & \\ & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6 & \\ & -1 \end{bmatrix} \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$